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Baseline risk factors for liver cancer development in a large multisite chronic HBV electronic health record-based cohort in the UK

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PURPOSE / OBJECTIVES

Nearly 300 million people worldwide are chronically infected with hepatitis B virus (HBV), which is the leading cause of global hepatocellular carcinoma (HCC) deaths. Although some risk factors for progression of chronic HBV infection (CHB) to HCC have been identified, such as male sex and increasing age, others have not been well characterised.

MATERIAL & METHODS

We conducted a retrospective cohort study including CHB patients from QResearch, a primary-care database of patient records from 2530 general practices in England (1), to identify individuals aged ≥ 18 years with CHB between 1999 and 2019. A univariable logistic regression (LR) model was fitted for each characteristic at baseline (the first record of CHB), with HCC as the primary outcome. A multivariable LR model was built using factors that were significantly associated with odds of HCC in univariable associations and/or identified a priori based on previous literature, using forward stepwise modelling with the Akaike information criterion.

RESULTS

We identified 7952 CHB patients (61% male) in whom 211 incident HCC cases occurred over a mean follow-up of 7.3 years. At baseline, 74% of patients were aged < 45 years, 44% were residing in the most deprived geographical regions, and > 60% were non-White.

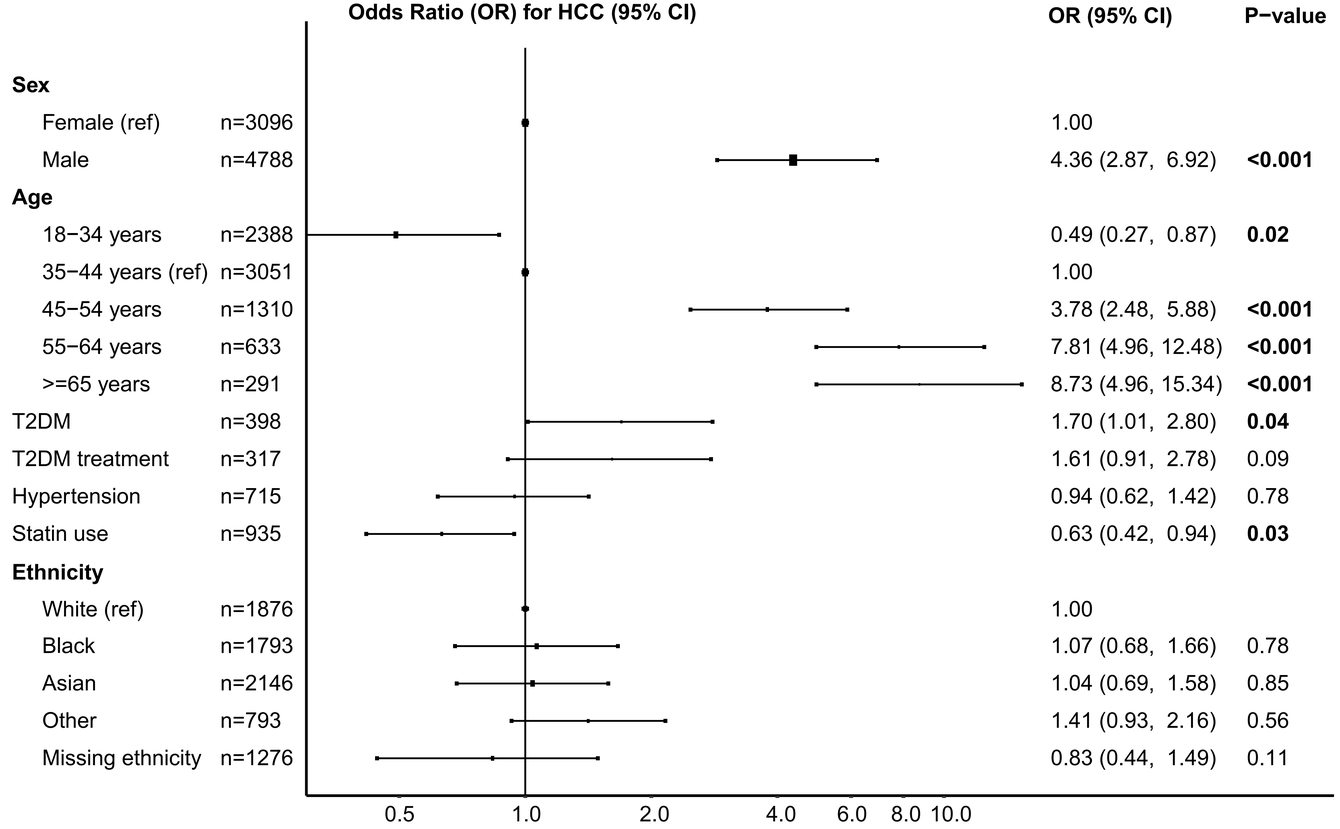

Figure 1. Odds ratios of hepatocellular carcinoma for baseline patient characteristics, estimated from a multivariable logistic regression model. Effect estimates are adjusted for all factors in the forest plot, in addition to socioeconomic status and body mass index.

1. **We characterised a young, deprived, majority non-White cohort of CHB individuals from primary care electronic health records**
2. **T2DM at baseline was significantly associated with increased odds of HCC**
3. **Statin use at baseline significantly associated with reduced odds of HCC**

RESULTS

In the multivariable model (adjusted for all factors in the forest plot plus socioeconomic status and body mass index, **Figure 1**), male sex and increasing age were significantly associated with increased odds of HCC, in line with previous literature. Baseline type 2 diabetes mellitus (T2DM) was significantly associated with HCC, with odds of HCC 70% higher (odds ratio (OR) 1.70, 95% confidence interval (CI) 1.01-2.80) in those with T2DM as compared to those without, after adjustment for additional confounders and T2DM treatment. Statin use at baseline was significantly associated with a 37% reduction in odds of HCC (OR 0.63, 95% CI 0.42-0.94) as compared to those who did not use statins.

SUMMARY / CONCLUSION

Our findings suggest that T2DM and other metabolic characteristics may be important prognostic risk factors for the development of HCC in CHB. Analysis is limited by gaps in the dataset such that significant risk factors for HCC may be overlooked and associations may be residually confounded.

ACNKOWLEDGEMENTS

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1. QResearch. Available at https://www.qresearch.org/.